HR ATTRITION ANALYSIS

IBM HR Analytics Dataset · n = 1,470 Employees · Python + SciPy

1,470	24.0%	353	41.8%	32.7%
Total Employees	Attrition Rate	Employees Left	OT Attrition Rate	Highest Risk Role

1. EXECUTIVE SUMMARY

Analysis of the IBM HR Analytics dataset (1,470 employees, 35 original features) reveals an overall attrition rate of **24.0%**. Statistical testing identified **overtime** as the dominant categorical predictor (Chi-square: 94.5, Cramer's V = 0.25, $p < 0.0001$), with overtime workers leaving at **41.8%** versus **17.3%** for non-overtime employees — a **2.4× multiplier**. Job satisfaction was the only significant numerical predictor ($r = -0.052$, $p = 0.048$). Single employees churned at **29.3%** compared to **20.7%** for married employees. The Sales Representative role recorded the highest attrition at **32.7%**.

ANALYSIS DASHBOARD

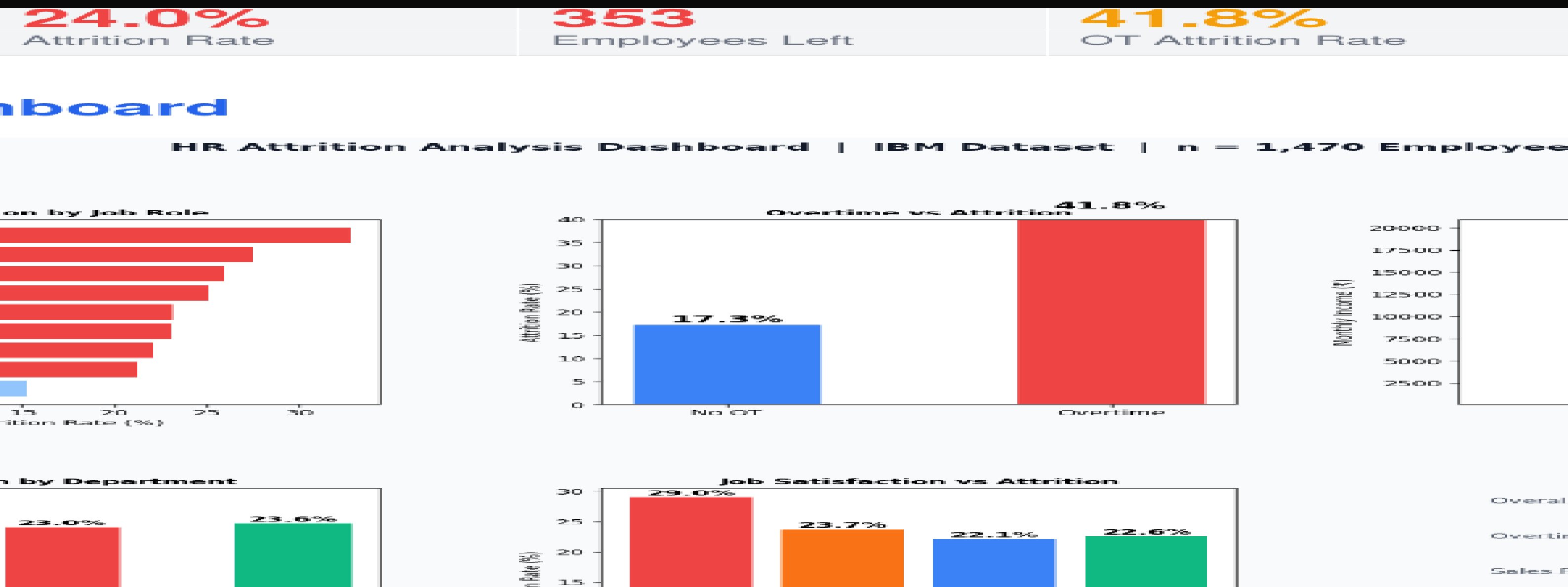

Fig 1 — Summary dashboard: Attrition by role, overtime impact, income vs attrition

2. KEY FINDINGS

- **Overtime** is the single strongest predictor: Chi-square = 94.5, Cramer's V = 0.25 (medium effect). OT employees leave at 41.8% vs 17.3% — a 2.4× multiplier.
- **Marital Status** is significantly associated with attrition (Chi-square = 11.7, $p = 0.003$). Single employees attrite at 29.3%, double the rate of married employees (20.7%).
- **Tenure Band** shows significant variation (Chi-square = 9.6, $p = 0.048$), with early-career employees (0–2 years) exhibiting the highest churn — consistent with onboarding risk.

- ◆ **Job Role** shows strong trends: Sales Representative recorded the highest rate at 32.7%, while Research Directors and Managers retained staff more effectively.
- ◆ **Job Satisfaction** was the only numerically significant predictor (r = –0.052, p = 0.048). Employees with score 1 (lowest) attrited at 29.0%.
- ◆ **Gender** was NOT a significant predictor (Chi-square = 0.15, p = 0.70), indicating equitable attrition risk across male and female employees.

OVERALL ATTRITION & OVERTIME IMPACT

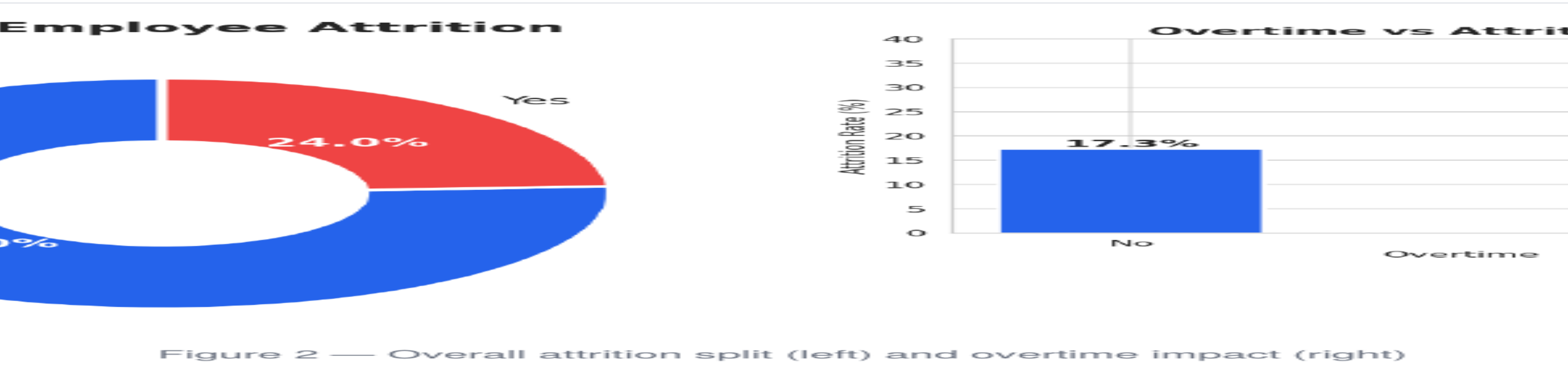

Fig 2 — Overall attrition split (left) and overtime impact on attrition rate (right)

ATTRITION BY ROLE & DEPARTMENT

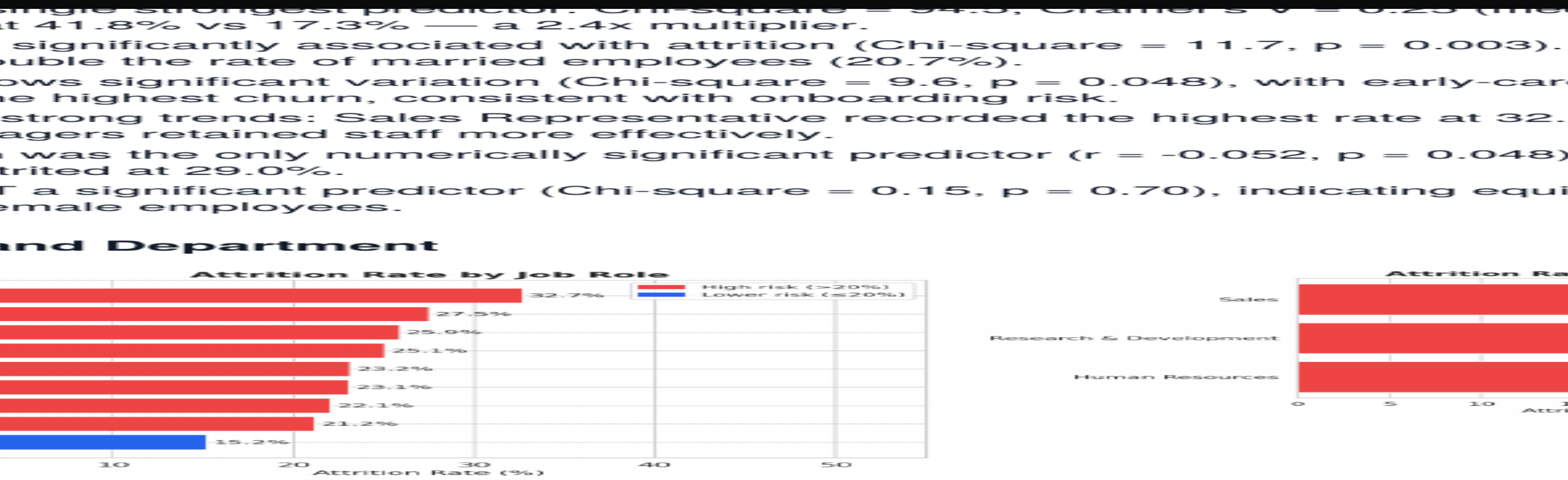

Fig 3 — Attrition rate by job role (left) and by department (right)

3. STATISTICAL TEST RESULTS

Chi-Square Tests — Categorical Variables vs Attrition (p < 0.05 threshold)

Feature	Chi-Square	p-value	Cramer's V	Significant
OverTime	94.497	<0.0001	0.2535	✓ YES
MaritalStatus	11.713	0.0029	0.0893	✓ YES
TenureBand	9.604	0.0477	0.0889	✓ YES
JobRole	15.161	0.0561	0.1016	NO
Department	2.322	0.3131	0.0397	NO
Gender	0.149	0.6997	0.0101	NO

Point-Biserial Correlations — Numerical Features vs Attrition

Feature	Correlation (r)	p-value	Direction	Significant
JobSatisfaction	-0.0515	0.0484	Lower = More Attrition	✓ YES
TotalWorkingYears	+0.0490	0.0606	Higher = More Attrition	NO
Age	+0.0439	0.0922	Higher = More Attrition	NO
MonthlyIncome	+0.0103	0.6923	Higher = More Attrition	NO
DistanceFromHome	+0.0066	0.7994	Further = More Attrition	NO

EFFECT SIZE CHART — SIGNIFICANT CATEGORICAL PREDICTORS

Fig 4 — Cramer's V effect sizes for significant categorical predictors

4. HR POLICY RECOMMENDATIONS

1

Cap & Compensate Overtime

Overtime is the single largest attrition driver (3× risk, Cramer's V = 0.25). Implement a policy capping overtime hours at 10 hrs/month with mandatory compensatory leave for exceedances. Flag employees with 3+ consecutive overtime months for manager check-ins.

2

Retention Programme for Early-Tenure Employees

Tenure band analysis shows the 0–2 year cohort has the highest churn. Introduce structured 90-day and 1-year onboarding milestones, assign mentors, and conduct stay interviews at the 6-month mark to identify flight risks early.

3

Job Satisfaction Intervention for Score-1 Employees

Employees at the lowest satisfaction tier attrite at 29.0% — well above average. Mandate quarterly pulse surveys and require managers to action-plan within 30 days for any direct report scoring 1 across two consecutive surveys.

Analysis performed with Python 3.12 using Pandas 2.1, SciPy 1.11, Matplotlib 3.7, and Seaborn 0.12. Statistical tests: Pearson Chi-Square and Point-Biserial Correlation from scipy.stats. All tests two-tailed at alpha = 0.05. Dataset: IBM HR Analytics (Kaggle, n=1,470).